

The Impact of Water Use Fees on Dispatching and Water Requirements for Water-Cooled Power Plants in Texas

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S Supporting Information

ABSTRACT: We utilize a unit commitment and dispatch model to estimate how water use fees on power generators would affect dispatching and water requirements by the power sector in the Electric Reliability Council of Texas' (ERCOT) electric grid. Fees ranging from 10 to 1000 USD per acre-foot were separately applied to water withdrawals and consumption. Fees were chosen to be comparable in cost to a range of water supply projects proposed in the Texas Water Development Board's State Water Plan to meet demand through 2050. We found that these fees can reduce water withdrawals and consumption for cooling thermoelectric power plants in ERCOT by as much as 75% and 23%, respectively. To achieve these water savings, wholesale electricity generation costs might increase as much as 120% based on 2011 fuel costs and generation characteristics. We estimate that water saved through these fees is not as cost-effective as conventional long-term water supply projects. However, the electric grid offers short-term flexibility that conventional water supply projects do not. Furthermore, this manuscript discusses conditions under which the grid could be effective at "supplying" water, particularly during emergency drought conditions, by changing its operational conditions.



■ INTRODUCTION

Scarce water supplies caused negative economic and social impacts across the State of Texas in 2011,¹ and expected climate changes and growth are expected to intensify these impacts.² The Texas Water Development Board (TWDB) administers a comprehensive state water plan (SWP) every five years that recommends water supply projects intended to meet 50 years of demand assuming drought of record conditions. In the 2012 SWP, the TWDB recommended investing more than 53 billion USD in traditional water supply projects (i.e., transmission, treatment, and new groundwater and surface water supplies), unconventional supplies (i.e., reuse, desalination, and conservation), and reallocation of existing supplies.³

Despite the significant water requirements of the power sector, water conservation strategies pertaining to Texas' electricity grid are not addressed in the TWDB's SWP. This analysis investigates potential reductions in water consumption and withdrawals by the power sector that would be triggered by an increase in the water cost paid by power producers.

■ BACKGROUND

Water use for electricity production in Texas is estimated at 2.5–4.2% of its annual water consumption^{4,5} and 40–65% of its annual water withdrawals.^{5,6} While Texas' population is projected to grow 84% between 2010 and 2060, water consumption for thermoelectric power generation is projected to grow 121% during this period, outpacing average population growth.³

Analysis of water "use" for power generation is complicated by different conceptualizations of "use". Withdrawn water is typically defined as the volume of water removed from a source water reservoir that might or might not be returned after use. Consumed water refers to the subset of withdrawn water that evaporates or is otherwise not returned to its original source and/or watershed.⁷ Water withdrawals are generally used in

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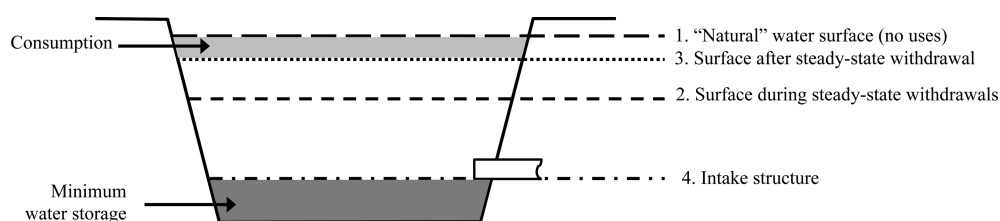


Figure 1. Schematic of a hypothetical cooling water reservoir illustrates anthropogenic withdrawals and consumption by an intake structure. The relative scale is exaggerated for the purposes of clarification.

water rights schemes, are more readily measured, and are thus used as a proxy for water “use”, even though much of that water might be simply passed through. This proxy can be misleading. Consider the hypothetical source water body shown in Figure 1. Absent any anthropogenic flows, stored water is at surface 1. During steady-state withdrawals, water storage drops to surface 2. The difference between surface 1 and 2 is equivalent to water stored in any infrastructure actively using water, such as water stored in cooling towers, intake piping, or drinking water storage tanks. Such storage (the difference between 1 and 2) is likely to be negligible relative to the storage in source water bodies (e.g., rivers and reservoirs). If withdrawals cease, storage rises to surface 3, with consumption represented by the differences between levels 1 and 3.

In Texas, consumption-to-withdrawal ratios in commercial, industrial, domestic, irrigation, livestock, and mining end-use sectors are 0.20, 0.27, 0.42, 0.86, 1.0, and 1.0, respectively. For thermoelectric use this ratio is 0.03.^{8,9} Thus, a unit decrease in water withdrawn for power generation has a near negligible reduction on consumption compared to other end-use sectors, underscoring the potential for withdrawn water to be ambiguous when used as a proxy for “use”.

Inconsistent terminology across water-related administrative agencies further complicates effective decision-making. The Energy Information Administration (EIA) and the United States Geological Survey (USGS) use the terms “withdrawals” and “consumption” to describe water flows. The TWDB uses the term “demand,” which is not explicitly defined and is used inconsistently with respect to consumptive and nonconsumptive uses: consumption is used for power and steam generation, but we infer that withdrawals are used for other sectors based on comparisons with other databases. Further, water rights issued to power plants by the Texas Commission on Environmental Quality (TCEQ) may be listed as “diversions” and/or “consumption”, with no consistent relationship between the two allocations and some allocations are grossly different than what would be expected given reported cooling technologies.^{6,8–10}

The volume of water withdrawn or consumed for power generation varies according to cooling technology, fuel type, prime mover (i.e., the technology responsible for converting thermal energy to mechanical work), and prevailing meteorological conditions.^{4,5,11,11–15} The water requirements of thermoelectric power generators, which include fossil fuel (i.e., coal, natural gas, and in limited cases, petroleum), biomass, and nuclear generators, vary by orders of magnitude. Open-loop cooled or once-through cooled power plants withdraw large volumes of surface water sources but typically return the majority back to the original reservoir. While once-through cooled plants lose relatively less water to the environment through evaporation, large volumes of withdrawn water can negatively affect aquatic ecosystems.^{4,16,17} Closed-loop or

recirculating cooled power plants, withdraw relatively less water, but a large fraction of the withdrawn water is lost to evaporation. Thus, power plants affect water availability for other water users in Texas due to their large water requirements.

Previous studies cite varying impacts of once-through cooled plants on water availability to downstream users. Scanlon et al.⁵ suggest that the presence of a once-through cooled plant increases water availability to downstream users by means of less net consumption in the reservoir in comparison to a recirculating cooled plant with similar generation, whereas Stillwell and Webber¹⁸ demonstrate that retrofitting once-through cooling systems with recirculating cooling towers can reduce water withdrawals for power plants, freeing potential transfer of water rights. Therefore, there are trade-offs between once-through and recirculating cooling systems: recirculating cooled plants increase water consumption from a river basin, thereby undermining water supply reliability; once-through cooled plants are vulnerable to water shortages if sufficient volumes of water are not available for cooling, thereby undermining power reliability. For power generators, a water shortage might cause (1) interruptions to power production in the case that a power plant does not have sufficient access to cooling water or (2) interruptions to the water availability to downstream users due to scarcity.⁴

Power generation technologies exist that require very little water. Combustion-turbine and open-cycle natural gas generation units often require no cooling water, as they are air-cooled, although newer versions use a minor amount of water to prechill air at the inlet of the turbine.⁵ These units are typically smaller in scale and are more expensive to operate but have fast ramp times, and are, therefore, usually procured for peak times and ancillary services. Combined-cycle plants generally combine two combustion turbines with one steam turbine, reducing cooling water requirements by two-thirds in comparison to plants using steam turbines alone. Although dry cooling systems (i.e., cooling systems that use air rather than water to cool hot steam) exist, these systems typically have large capital costs and reduce power plant efficiency.⁴ Wind and solar photovoltaic systems require no water for cooling but are constrained by the availability of wind and solar resources and cost.¹¹

The Electric Reliability Council of Texas (ERCOT) is responsible for managing and operating the electric grid across the majority of Texas. Former analyses have investigated various energy-water management strategies to reduce the water intensity of power production within ERCOT. Stillwell and Webber¹⁶ conclude that increasing water storage at the site of power production increases the reliability of the power plant but has detrimental effects on downstream water users. Pacsi et al.¹⁹ analyze the potential for environmental dispatching (i.e., dispatching EGUs according to water availability) to reduce

Table 1. Marginal costs (i.e. the sum of fuel costs and operational and maintenance costs (O&M)) and water requirements of 310 ERCOT EGUs (Plus Aggregated Wind Generators) are characterized into 12 categories based on fuel type (FT), cooling technology (CT), and prime mover (PM)^a

EGU technology	EGU count	capacity range	avg. fuel cost	avg. O&M cost	avg. marginal cost	consumption rate		withdrawal rate	
						avg. cons. rate	std. deviation	avg. withdrawal rate	std. deviation
FT-CT-PM	(units)	(MW)	(\$ MWH ⁻¹)	(\$ MWH ⁻¹)	(\$ MWH ⁻¹)	(gal MWH ⁻¹)	(gal MWH ⁻¹)	(gal MWH ⁻¹)	(gal MWH ⁻¹)
BM (Varied)	7	4–100	31	10	41	35	51	376	553
WI-NA-WI	1	NA ^b	0	0	0	0	0	0	0
HY-NA-HY	30	3–40	0	0	0	0	0	0	0
NGIC-NA-CT	31	6–10	40	3	43	44	15	44	15
OCGT-NA-CT	79	13–437	56	4	60	10	7	10	7
NG-OT-ST	46	20–742	45	7	52	13	43	1216	3826
NG-RC-ST	10	24–139	43	6	49	42	108	195	571
NG-OT-CC	3	186–706	35	4	39	37	65	6548	12 154
NG-RC-CC	57	10–1431	33	3	36	101	39	289	258
NG-DR-CC	10	231–241	32	5	37	100	0	100	0
CL-OT-ST	22	443–811	18	5	23	373	62	29 198	3260
CL-RC-ST	11	157–854	18	5	23	564	14	752	22
NU-OT-ST	4	1179–1363	5	4	9	584	0	120 000	0

^aValues reflect generation weighted averages in the 2011 baseline scenario. (All wind generators are aggregated into a single unit and dispatched to match 2011 wind generation.). *Fuel Type*: nuclear (NU), coal (CL), natural gas (NG), biomass (BM), wind (WI), hydro (HY), open-cycle natural gas (OCGT) and internal-combustion natural gas (NGIC). *Cooling Technology*: once-through (OT), recirculating (RC), dry-cooled (DR), no cooling (NA). *Prime Mover*: steam boiler (ST), combined-cycle (CC), combustion turbine (CT), wind turbine (WI), hydro turbine (HY). ^bWind Generators are aggregated into a single unit and collectively have a 3.5 GW average hourly annual operating capacity.

water competition between power producers and other users during scarcity events. They conclude that shifting electricity production from power plants in drought-stricken regions of South Texas during the 2006 drought would have been feasible in the context of ERCOT's transmission and distribution constraints. Others have evaluated the use of alternative cooling technologies or water sources;⁴ however, to the authors' knowledge, increased valuation of water through market levers as a mechanism to induce water savings in the power sector has not been analyzed. This paper seeks to analyze that missing element.

The generation and dispatching of electricity within ERCOT is governed by a unit commitment and dispatch (UC&D) system. Such a system minimizes the marginal operating cost while meeting the electricity demands in ERCOT's service area by dispatching power production in order of least marginal cost. Although operation and maintenance costs reflect some portion of the cost of cooling water supplies, these costs are nearly negligible given Texas' historical water lease rate (i.e., 100 USD per acre-ft).⁴ Under current operating conditions, the majority of water withdrawn and consumed for power generation across ERCOT is associated with lower-cost generators, while the majority of the least water intensive generators are more expensive to operate and are utilized with low capacity factors under current grid operations (i.e., least marginal cost basis).²⁰ However, if the water for power plants incurred a higher cost, the order in which power generators are dispatched within ERCOT might shift toward water-lean generators. This analysis presents a methodology for investigating that effect by imposing a notional cost on water used by power producers in ERCOT.

MATERIALS AND METHODS

A UC&D model of ERCOT (detailed in the Supporting Information (SI) document) was modified to consider water use characteristics for 310 ERCOT Electric Generating Units

(EGUs) (plus aggregated wind generators) in 2011. The UC&D model was implemented in Energy Exemplar's PLEXOS for Power Systems (PLEXOS) version 6.208. PLEXOS simulates competitive wholesale power markets by means of linear, mixed integer and quadratic optimization that minimizes grid-wide short-run marginal operating costs required to meet demand. The short-run marginal cost (SRMC), defined in eq 1, includes variable operations and maintenance costs, fuel costs, and environmental costs. PLEXOS, thus, allows grid simulation of environmental flows currently not reflected in the market.

$$\text{SRMC}_{\text{EGU}} = C_{\text{VO\&M}} + C_{\text{fuel}} \times \text{MHR} + C_{\text{enviro}} \times Q_{\text{enviro}} \quad (1)$$

In eq 1, variable operation and maintenance costs, $C_{\text{VO\&M}}$, are in units of USD per MWh; fuels cost, C_{fuel} , is in USD per million British Thermal Units (MMBTU); marginal heat rate, MHR, is in MMBTU per MWh; the environmental cost, C_{enviro} , is in USD per unit volume of water (zero under baseline conditions); and the withdrawal or consumption rate, Q_{enviro} , is in terms of consumed (or withdrawn) volume per MWh generated. In our study, we use ERCOT's 2011 load profile,²⁰ fuel pricing,^{21–24} variable operating and maintenance costs,^{25,26} and EGU-specific water use characteristics to calibrate a UC&D model reflective of 2011 conditions. We then characterize changes to grid performance assuming costs for water consumption alone or the entire volume of water withdrawn for cooling.

Water consumption rates were assigned for each EGU in ERCOT based on a report prepared by King et al.²⁷ for the TWDB based on data collected by the Texas Commission on Environmental Quality (TCEQ). Water withdrawal rates are not measured by the TWDB or TCEQ, so we used data reported in the 2010/2011 EIA-923 forms¹⁰ and the Union of Concerned Scientists' EW3 Energy-Water database documented by Averyt et al.²⁸ Data sets were checked for

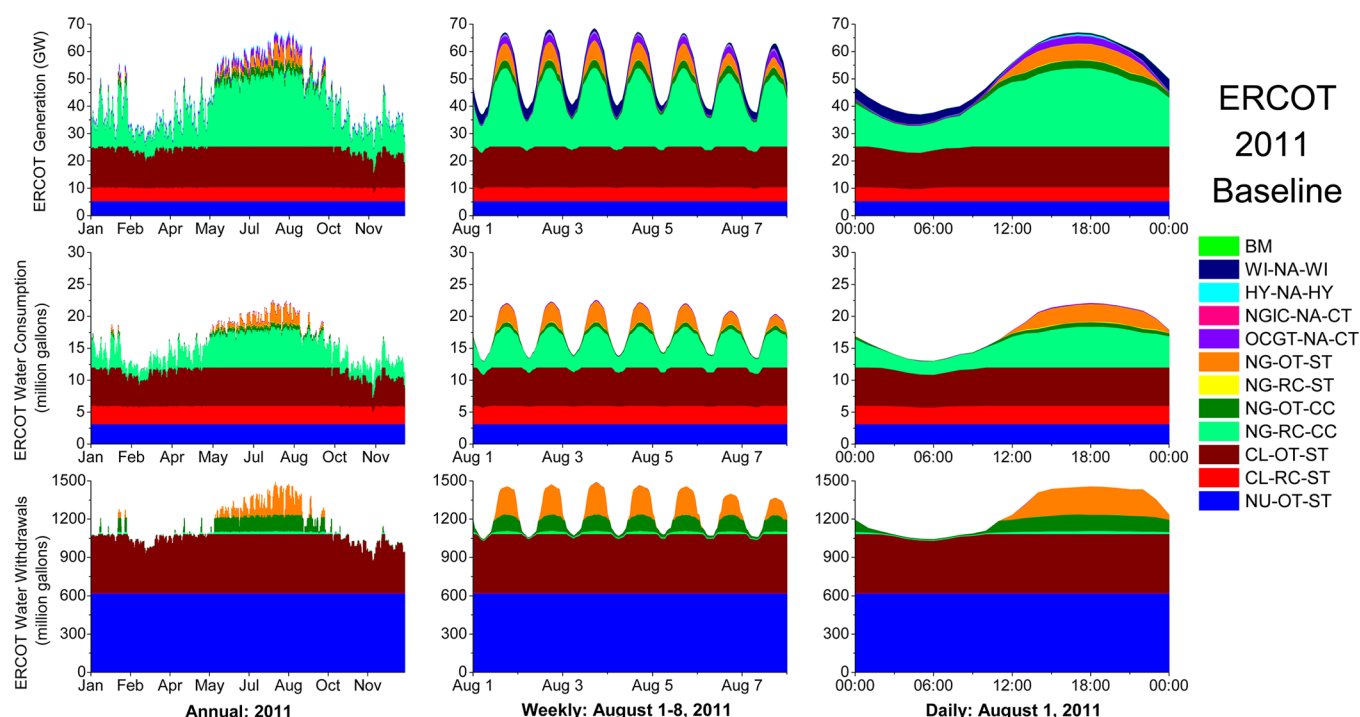


Figure 2. Baseline power generation in ERCOT in 2011 was simulated using historical fuel prices and the 2011 load profile from that year. Nuclear and coal EGUs provide steady baseload across the year, whereas natural gas combined-cycle plants ramp up and down to meet variable daily and seasonal loads. (The 10 NG-DR-CC units are included in the NG-RC-CC category here since their aggregated generation is very low.)

thermodynamic plausibility and cross-checked for accuracy. (See SI for additional details.)

Table 1 summarizes generation-weighted average water use rates for 310 EGUs (plus aggregated wind generators) in ERCOT by fuel type, cooling technology, and prime mover technology combination in the 2011 baseline scenario. Hydroelectric facilities were assumed to have no water consumption and no withdrawals for power production, although in reality the presence of a dam increases evaporation relative to the natural run of the river (i.e., without a dam).²⁹ Overall, nuclear generators withdrew and consumed the most water of any other type of generator. Generally, pond-cooled and once-through cooled EGUs had the highest withdrawal rates across any cooling technology. EGUs with recirculating cooling averaged higher water consumption rates when compared against EGUs with once-through cooling within the same fuel and prime mover technology categories.

A baseline scenario was analyzed to verify that water use characteristics in the 2011 baseline simulation were consistent with historical water use. Generation fleet characteristics and total wholesale generation costs were consistent with 2011 ERCOT operations. Annual water withdrawals in the final calibrated 2011 baseline UC&D model were estimated to be 9.3 trillion gallons per year; of that total, approximately 118 billion gallons were consumed in the simulation. The withdrawal estimate agrees within 10% with a recent analysis by Scanlon et al.,⁵ which concludes that withdrawals in Texas for thermoelectric power generation exceeded 8.5 trillion gallons in 2010. Scanlon et al.⁵ estimate 2010 water consumption in ERCOT to be 140 billion gallons, approximately 19% higher than the baseline.

After the baseline UC&D model was calibrated, dispatch scenarios were defined assuming increasing marginal water fees for generation. The first set of water cost scenarios considered

the effect of increasing the cost of water to EGUs based only on the volume of water *consumed* for power generation in 2011; the second set of scenarios applied the increased water cost to total volume of water *withdrawn* for power generation. Water consumption cost scenarios are referred to with “WC” and the water cost in USD per acre-ft (i.e., WC10, WC100, and WC1000 for increased costs of 10, 100, and 1000 USD per acre-foot, respectively). Withdrawal cost scenarios are referred to with “WW” and follow the same convention. These costs were chosen because they could be readily comparable to historical lease rates (i.e., 100 USD per acre-ft) or the order-of-magnitude costs associated with developing new water supply projects defined in the TWDB’s SWP (cost ranges provided in Table 3).³

We calculate the cost-effectiveness (CE) of “supplying” water through the power grid using eq 2. In eq 2, the denominator specifies the water that would be conserved (i.e., “supplied”) through the added marginal power generation cost, ΔMC_{gen} , due to increased water fees. We assume the water stored during generation (V_{store} , the difference between surfaces 1 and 2 in Figure 1) is negligible; thus CE simplifies to marginal cost of generation divided by the water reduction in consumption (ΔV_c).

$$CE = \frac{\Delta MC_{\text{gen}}}{V_{\text{store}} + \Delta V_c} \quad (2)$$

RESULTS

Water Use Results. Figure 2 summarizes baseline 2011 electricity generation (first row), water consumption (second row), and water withdrawals (third row) in ERCOT on an annual (first column), weekly (second column), and daily (third column) basis. EGUs are categorized by fuel, cooling technology, and prime mover. In baseline scenario, nuclear

Table 2. Raising the cost of water to EGUs across ERCOT induced varying levels of water reductions. Increasing the cost of water withdrawals incurred larger water reductions than increasing the cost of consumed water alone, but also incurred larger costs

scenario identifier	annual generation (TWh)	annual water costs (billion USD)	annual total generation cost ^a (billion USD)	annual water consumption (trillion gallons)	change in annual consumption	annual water withdrawals (trillion gallons)	change in annual withdrawals
Baseline	335	0	8.25	0.118		9.33	
WC10	335	0.00362	+0.21%	0.118	0.0%	9.33	0.0%
WC100	335	0.0362	+0.60%	0.118	−0.1%	9.32	−0.1%
WC1000	335	0.358	+4.5%	0.117	−1.3%	9.28	−0.6%
WW10	335	0.283	+3.6%	0.118	−0.5%	9.21	−1.3%
WW100	335	1.69	+28%	0.105	−11.3%	5.51	−40.9%
WW1000	335	7.34	+120%	0.0906	−23.4%	2.39	−74.4%

^aTotal Generation Cost (wholesale) includes includes total fuel, VO&M, start-up and shut-down, and water costs for EGUs across ERCOT.

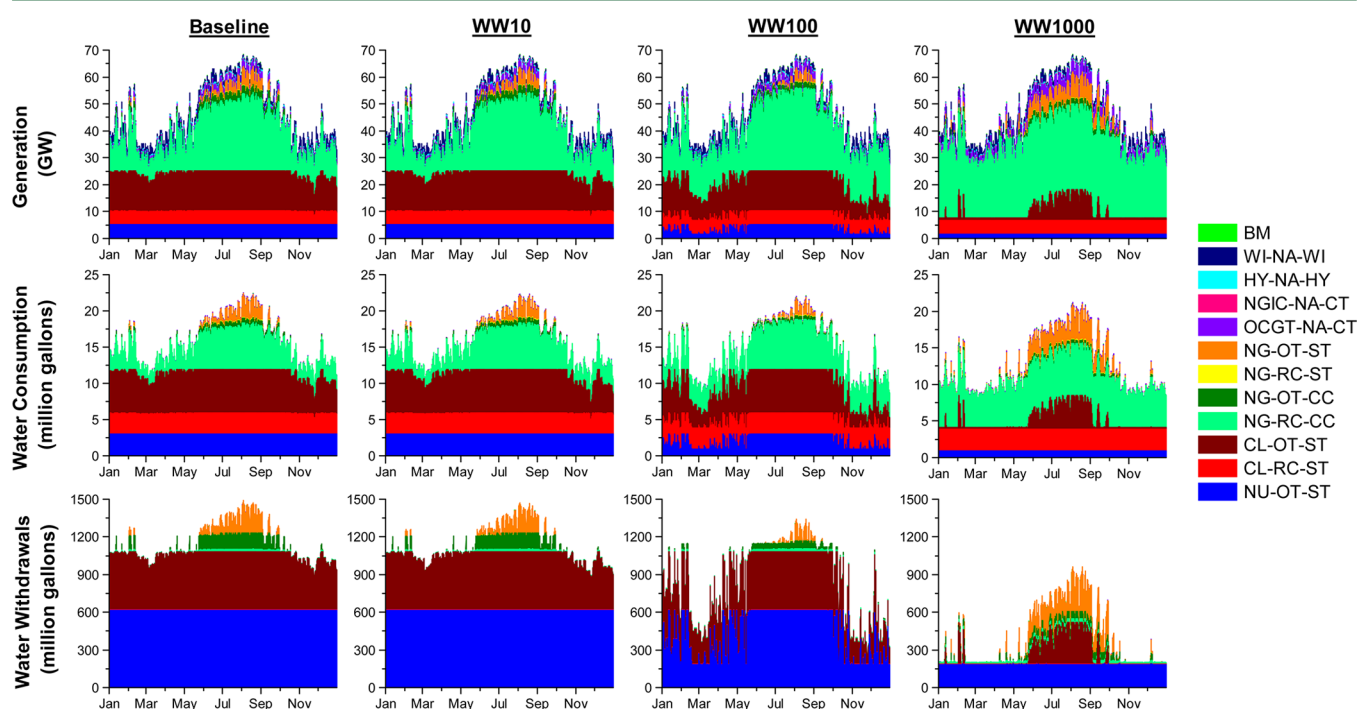


Figure 3. ERCOT-wide generation, water consumption, and water withdrawals are detailed for the baseline, WW10, WW100, and WW1000 scenarios. Generation by natural gas combined-cycle EGUs with recirculating cooling increases as the cost of water withdrawals increase. Once-through cooled EGUs have large decreases in annual generation as these costs increase, resulting in significant reductions in water withdrawals across ERCOT. (The 10 NG-DR-CC units are included in the NG-RC-CC category here since their aggregated generation is very low.)

EGUs operated at a steady capacity of approximately 5 GW for the majority of the year. Coal EGUs with recirculating cooling also operated at a relatively steady capacity, averaging 4.5 GW over the year. Coal EGUs with once-through cooling also served as steady baseload capacity but did exhibit some ramping up and down, especially in mild months when electricity demand was low (e.g., February, April, October, and November). Natural gas combined-cycle plants with recirculating cooling represented nearly 30% of total annual ERCOT generation in the baseline case, but exhibited daily ramping according to fluctuations in demand. These combined-cycle units provided as much as 28 GW of operating capacity during the hours of highest demand in 2011, and as little as 8 GW during hours of low load. Natural gas plants with steam boilers typically only operated in summer afternoons when demand was the highest due to very high marginal cost.

The results for the seven scenarios (including the baseline) are provided in Table 2. Results suggest that the water use

reductions following increases in the cost of water to power generators are nonlinear and vary according to whether the increased cost is applied to the subset of consumed water (WC scenarios) or the entire volume of withdrawn water (WW scenarios). Total wholesale generation costs increase from the baseline due to (1) increased water costs and (2) increased generation costs (independent of water costs) since the generation fleet and dispatch order shifts from the optimal cost baseline toward water-lean (but more expensive) generation.

Water withdrawal and consumption reductions for WC scenarios were modest compared to WW scenarios. Figure 3 details the generation (top row), water consumption (middle row), and water withdrawals (bottom row) of the baseline, WW10, WW100, and WW1000 withdrawal cost scenarios (listed from left-most to right-most column, respectively). Large shifts in generators occur when water costs are 100 USD per acre-ft and more. WC scenarios are not pictured, since

results indicated comparably smaller changes from the baseline. Additional details are in the SI.

Cost Effectiveness Results. The 2012 State Water Plan (SWP) recommends strategies that exceed a net present value of \$3B USD. The reported cost effectiveness of these strategies varies widely from around \$1 to \$60,000 per million gallons (\$0.33 to \$20,000 per acre-ft).⁸ Current and previous SWPs recommend water management strategies that meet future demands assuming the drought of record observed during the 1950s.

The 2011 drought challenged the current planning paradigm, putting over 90% of the state in extreme or exceptional drought and resulting in approximately \$7.6 billion of direct economic losses in the agricultural sector alone.¹ It is interesting to note that the additional generation costs in the WW1000 scenario were comparable to these costs, and reduced daily water consumption and water withdrawals by approximately 0.08 and 19 billion gallons per day, respectively. (To put this in context, the TWDB reports that the state required an average of approximately 4.5 billion gallons per day for municipal water uses in 2011.⁸)

Recommended water management strategies overwhelmingly rely on conventional water resource engineering “supply side” projects, such as developing new reservoirs and groundwater wells. While these projects have historically reduced the long-term impacts of drought, their development can take years if not decades. Moreover, they provide limited to no short-term flexibility to alleviate regional water scarcity. Due in part to this lack of flexibility, water users struggled for effective emergency responses during 2011.³⁰

Thus, we see a need for more effective, coordinated short-term responses to water scarcity, particularly in regions experiencing growth and climate change. The power grid does provide flexibility relative to conventional water supply projects in that (1) the infrastructure is currently in place to reduce consumption in the moderate- to short-term; (2) the grid is connected across watershed boundaries; (3) the power sector may be the only major water user that can substitute away from water consumption and maintain adequate services; and (4) reductions in water consumption do not require large capital outlays typical of conventional supply projects.

The CE results are shown in Table 3, which indicate that the cost effectiveness of water “supplied” through changes to grid operations range from about \$200,000 per million gallons to \$500,000 per million gallons. The WW100 and WW1000 scenarios provide total reductions in water consumption of 13 and 28 billion gallons per year, respectively. (For context, 1,460 billion gallons of new water supplies have been proposed to meet Texas’ demand in the year 2020.)

Table 3 indicates that power management through the grid is not as competitive with long-term conventional water supply projects on the basis of cost effectiveness. However, the grid provides flexibility that conventional water supply projects do not, as described above. Leveraging this flexibility would likely require weeks- to months-ahead planning based upon expected drought conditions to allow for the accumulation in storage in water bodies used for power generation. Similarly, coordination across appropriate state agencies would be needed to manage regional disparities in scarcity and respective variable changes to power grid operations. A more rigorous assessment of the CE of water “supplied” through the grid would assess such regional variation and reflect the probabilistic nature of drought,

Table 3. The cost effectiveness of reducing water consumption in Texas through the power grid is generally not competitive with conventional water management strategies

scenario	annual added generation cost from the baseline (million USD)	annual reduced water consumption (million gal)	cost effectiveness of reduced water consumption (\$/million gal)
WC10	17	35	486 000
WC100	49.4	165	300 000
WC1000	372	1530	244 000
WW10	294	596	493 000
WW100	2320	13 300	174 000
WW1000	9820	27 600	355 000
2012 TWDB State Water Plan		Projected Cost of New Water Supplies	
new supply strategies		(\$/acre-ft)	(\$/million gal)
all recommendations in SWP		0.33–20 000	1–60 000
new surface water supplies in SWP		65–5900	200–18 000
new groundwater supplies in SWP		16–1400	50–4300
reused/reclaimed water		29–1300	90–4000
new desalination supplies		160–1300	500–4000
bottled water		390 000	1 200 000

identifying hydrological conditions that render grid changes effective.

We also see value in using the power grid to free water from cooling system minimum water storage (MWS) as defined in Figure 1. This MWS (also referred to as “dead pool storage” by some agencies) can be defined by the elevation of the cooling system intake required to withdraw water as well as limitations on thermal discharges into cooling water bodies (e.g., Clean Water Act Section 316(a)). In essence, the existence of the MWS dictates that less than 100% of the total reservoir storage is available for release to other water users in the water basin. The MWS is site-specific, depending on the design of the intake and discharge structures as well as local climate. In many cases, the MWS is much greater than annual power plant water consumption, and thus freeing some of the MWS (e.g., by plant retrofit or shifting to other generators) might make grid operations more cost-effective in managing water. That analysis, however, is beyond the scope of this study and targeted for an area of future work.

While this analysis facilitates important considerations for broader water management strategies in Texas, it has several limitations. First, unclear and inconsistent terminology regarding water “use” as applied in the SWP likely affects a fair comparison of water supply alternatives on the basis of CE. Our analysis does not recognize any constraints imposed by current water rights nor did we consider the influence of hydrology and spatial variation in our results, which would recognize the added value of water saved at higher elevations or the impact on specific receiving water bodies. Similarly, we did not consider the economic implications of appropriating additional generation costs, or the issue of “who pays” for the added water use fees. The short-run own-price elasticity of electricity demand is around -20% ,³¹ which means passing added generation costs on to consumers would involve nontrivial consumer responses that would reduce electricity consumption (and respective water use). Such a response would be greatly diluted (likely unobservable) if the added generation costs were administered through general revenue streams (as are the TWDB projects). An economic equilibrium

approach is beyond the scope of this study, but we emphasize the importance of such economic considerations for future studies. Furthermore, this analysis restricted itself to the vintage fleet of 2011. It did not contemplate the impact of water fees on the economics of newly constructed power plants. It is expected, however, that these results will be informative to future planners contemplating that exact question.

■ ASSOCIATED CONTENT

● Supporting Information

Additional information as noted in the text. This material is available free of charge via the Internet at <http://pubs.acs.org/>.

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Notes

The authors declare no competing financial interest.

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